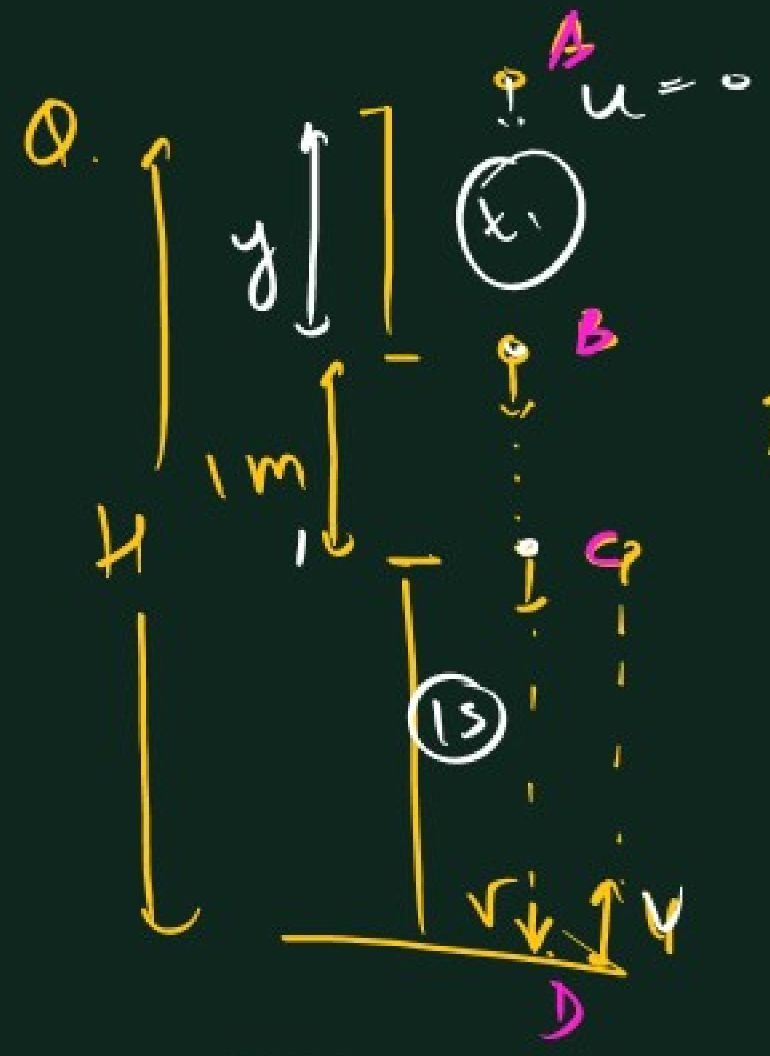
Q.



B to C = 0.1 s

$\boxed{C \to D \to C = 2s.}$

C to D → 1s.

$$s = \frac{1}{2}at^2$$

$$A \to C \Rightarrow y + 1 = \frac{1}{2} \times 10 \times (t_1 + 0.1)^2 \quad \text{—(1)}$$

$$A \to B \Rightarrow y = \frac{1}{2} \times 10 \times t_1^2 \quad \text{—(2)}$$

$$A \to D \Rightarrow H = \frac{1}{2} \times 10 \times (t_1 + 0.1 + 1)^2 \quad \text{—(3)}$$

(1) − (2)

$$1 = \frac{1}{2} \times 10 \times \left((t_1 + 0.1)^2 - t_1^2\right)$$

$$1 = 5 \times (2t_1 + 0.1)\, 0.1$$

$$\Rightarrow 2 = 2t_1 + 0.1$$

$$\Rightarrow t_1 = \frac{2 - 0.1}{2} = \frac{1.9}{2}$$

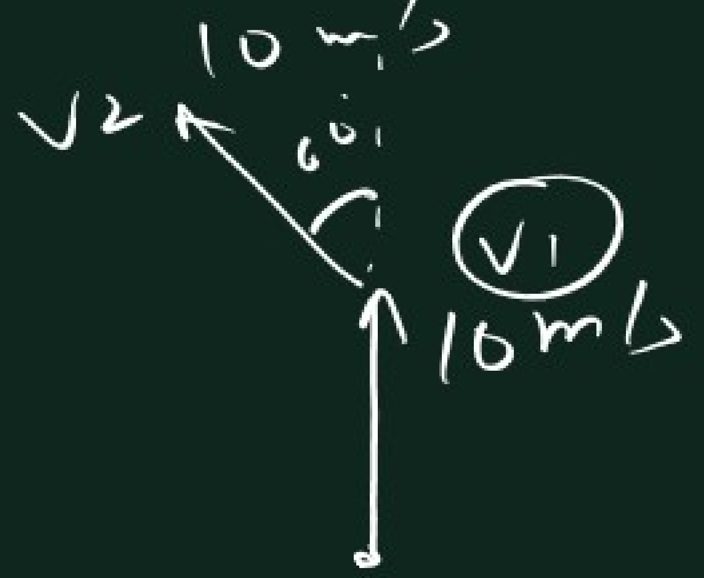
Put t_1 in (3)

$$H = 5\left(0.95 + 0.1 + 1\right)^2$$

$$= 5 \times (2.05)^2$$

$$= 21.0125$$

5)

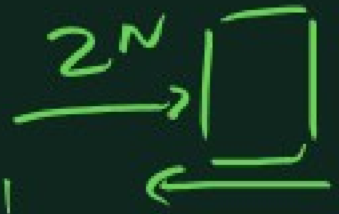


$$F = \frac{\Delta P}{\Delta t} = \frac{m\,\Delta v}{\Delta t} = \frac{m(v_2 - v_1)}{t}$$

$$= \cancel{m}\sqrt{10^2 + 10^2 - \cancel{2} \times 10 \times 10 \times \frac{1}{\cancel{2}}}$$

$$= 10$$

Friction

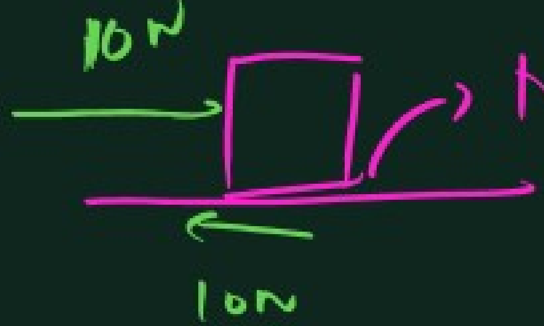


6000000...!

$f = 2N$

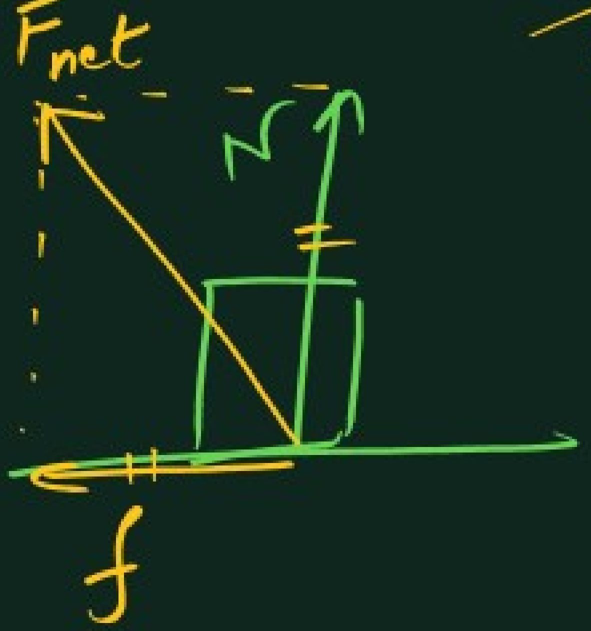
$f = 5N$

$f = 6N$



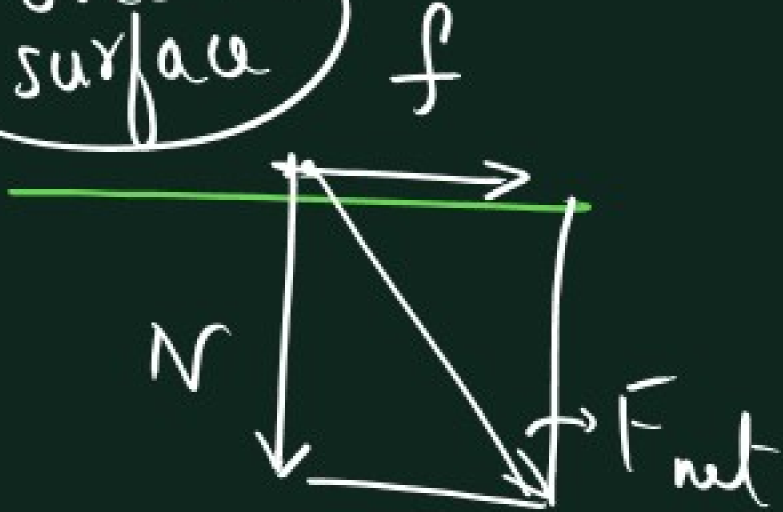
Max. $f = 20N$

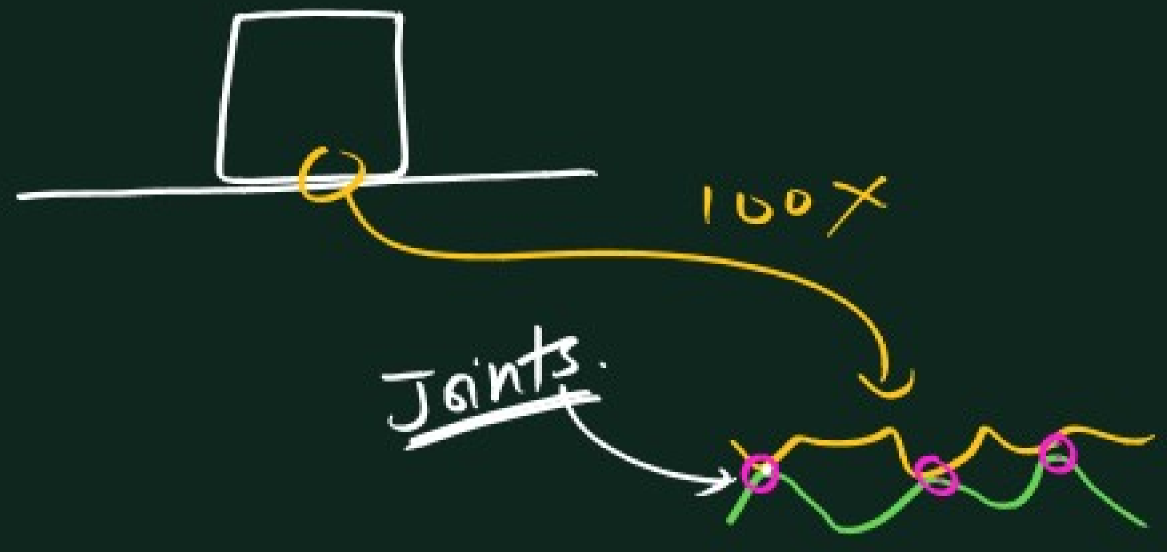
$f_{kinetic}$

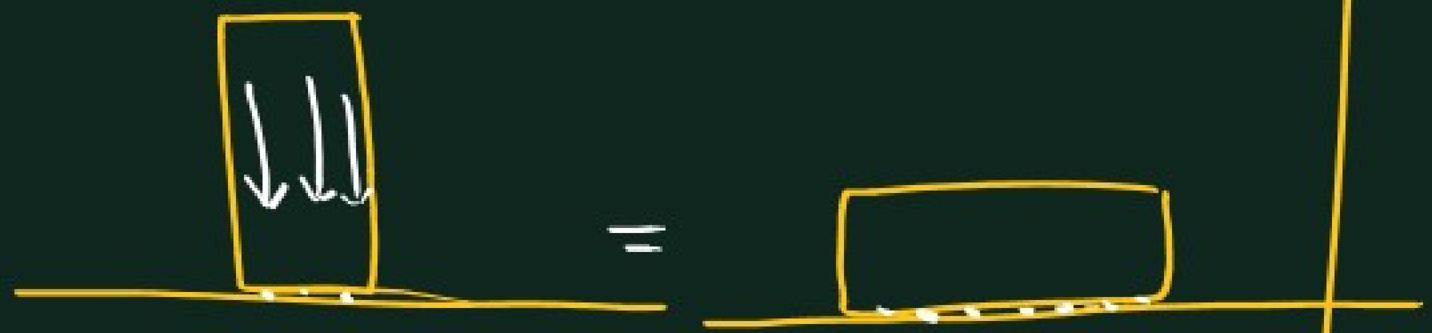


$$F_{net} = \sqrt{N^2 + f^2}$$

Force on surface







If $F = 0$, $f = 0$

$F = 2N$, $f = 2N$

$F = 5N$, $f = 5N$

$F = 6N$, $f = 6N$

$F = 6.0001 \Rightarrow$ Motion

$\Rightarrow f_{kinetic}$ acts.

$$\boxed{0 \leq f_{static} \leq f_{limiting}}$$

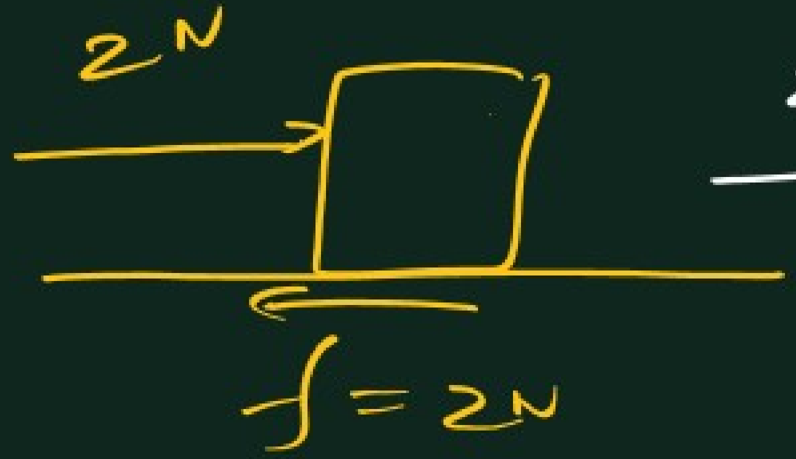
$f_{limiting}$ = max. possible value

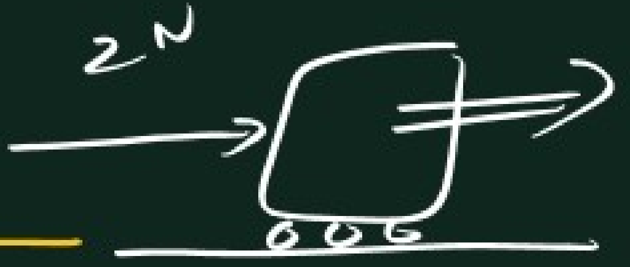
(f_L)

$$\boxed{f_{\underline{L}im.} = \mu_s N}$$

(Generally $\mu < 1$)

N = Normal Rxn.

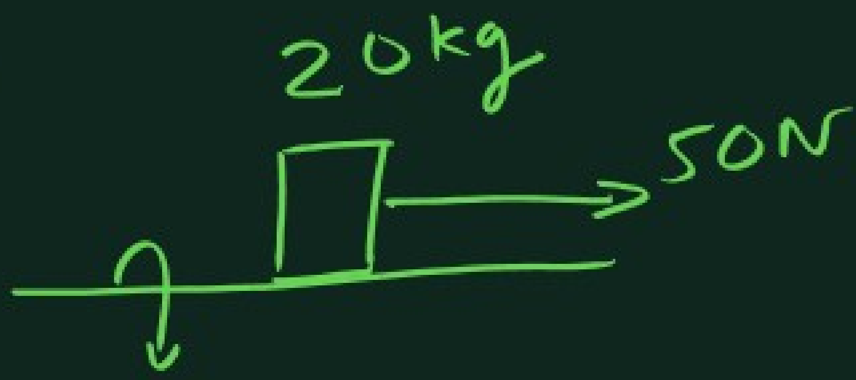




How to find tendency of motion

→ Assume $f = 0$ (wheels/oily)

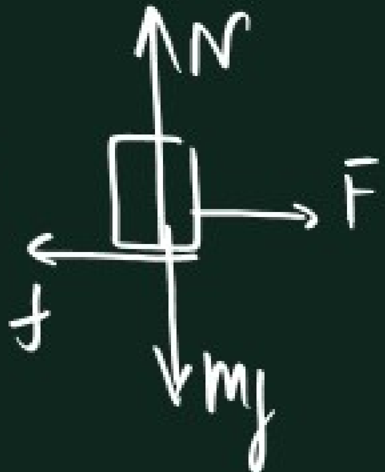
Q1.



$\mu = 0.5$

Find
1) limiting f
2) Static f acting

1)

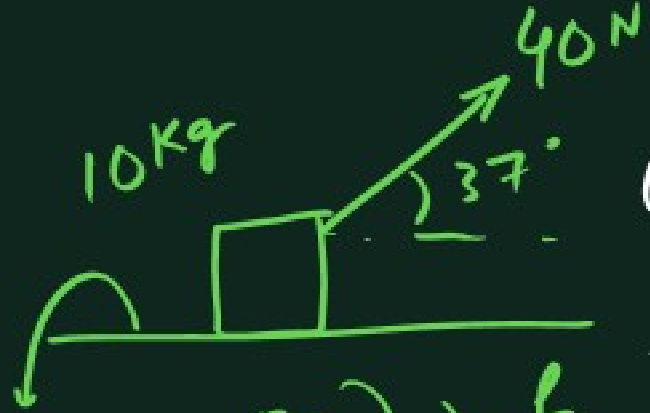


$N = mg = 200\,N$

$f_L = \mu N = 0.5 \times 200 = 100\,N$

2) f acting $= F_{external} = 50\,N$

2)

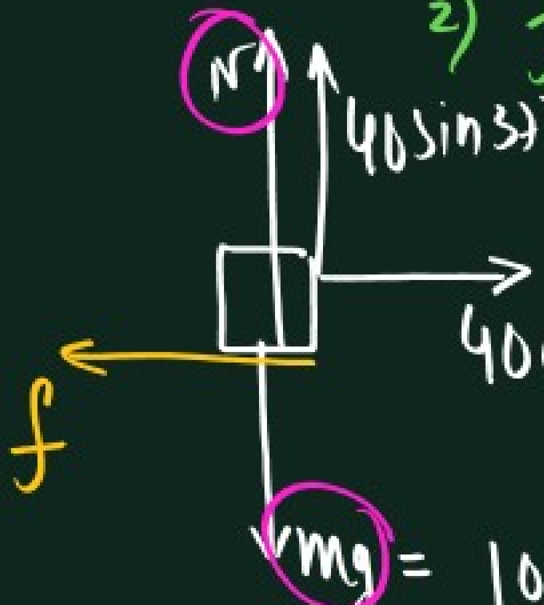


$\cos 37 = \frac{4}{5}$

$\sin 37 = \frac{3}{5}$

$\mu = 0.5$ Find 1) f_L

2) f acting



$40 \sin 37 = 24$

$40 \cos 37 = 32$

$mg = 100$

1) $\Sigma F_y = 0 \Rightarrow N + 24 = 100 \Rightarrow \boxed{N = 76}$

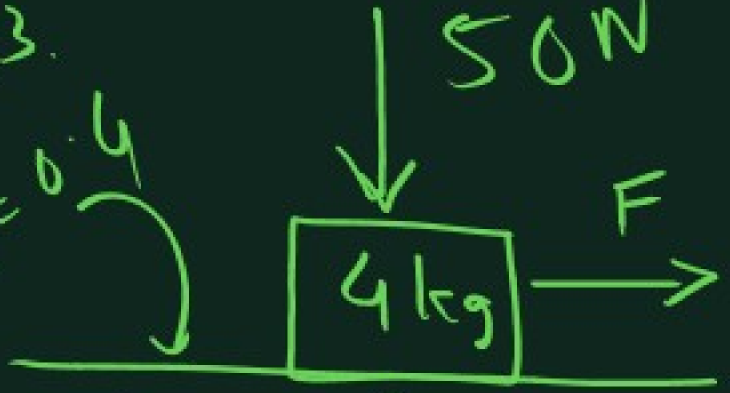
$f_L = \mu N = 0.5 \times 76 = 38\,N$

2) f acts opposite to net force parallel to surface

$f = 32\,N$

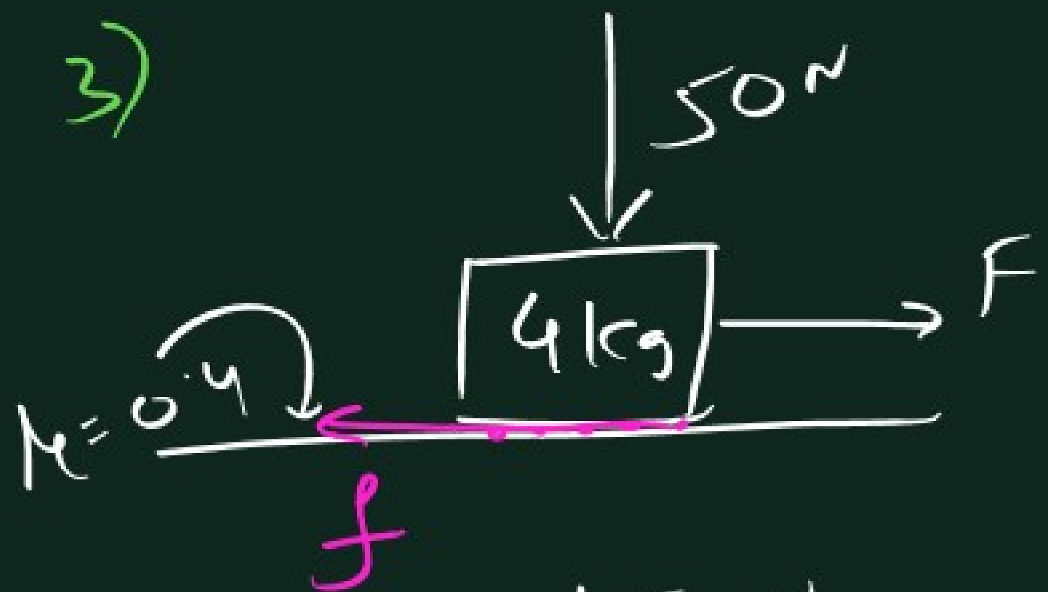
Q3.

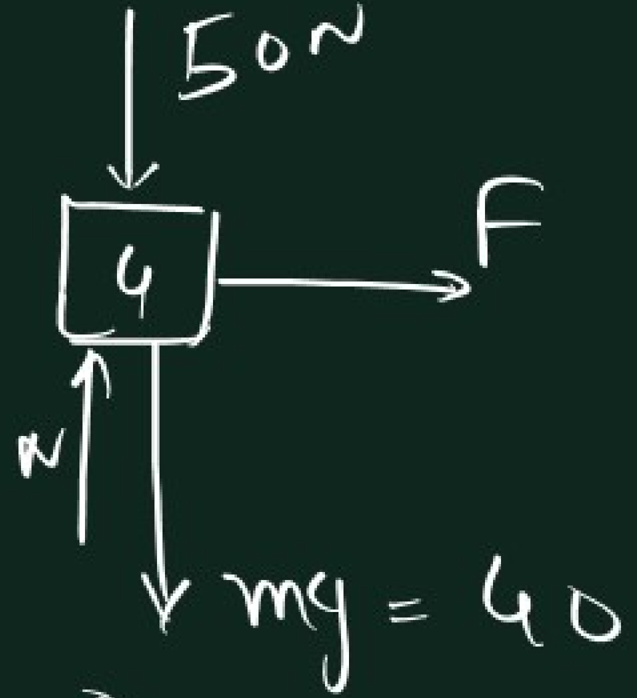
$\mu = 0.4$



Find 1) $f_{limiting}$
2) Minimum F to slide box

3)





$\Sigma F_y = 0$

$\Rightarrow N = 50 + 40 = 90$

1) $f_L = \mu N$

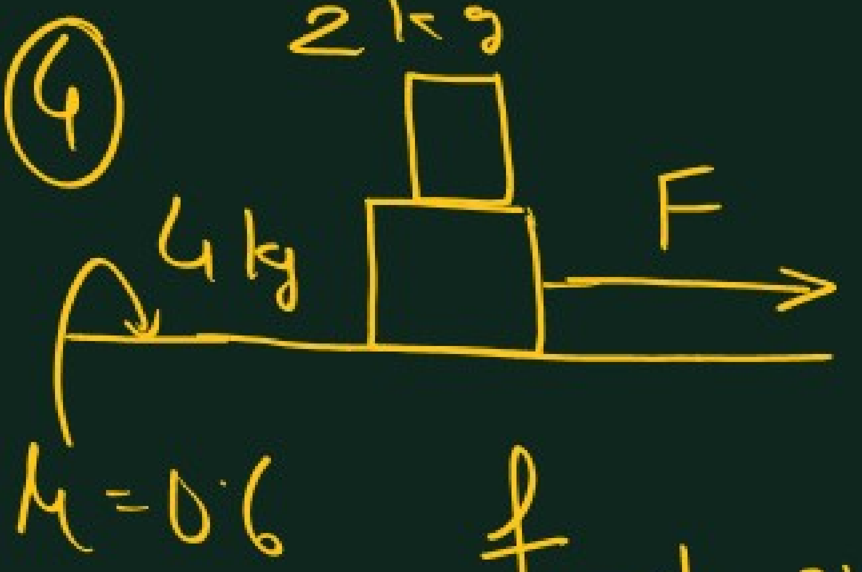
$= 0.4 \times 90$

$= 36N$

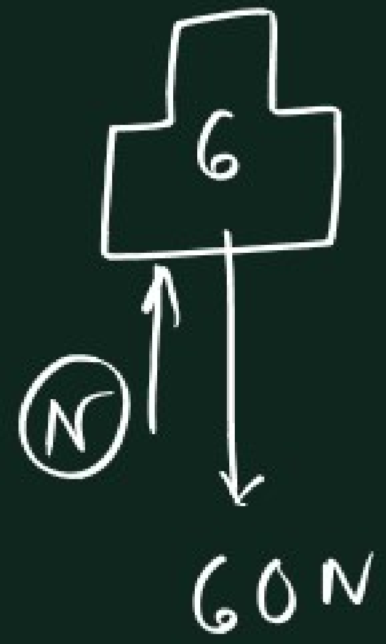
2) F_{min} to move $= f_L$

$= 36N$

4)



$f_{limiting}$?!



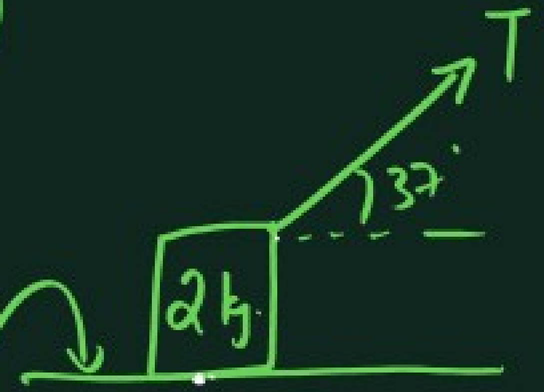
$N = 60$

$\therefore f_L = \mu N$

$= 0.6 \times 60$

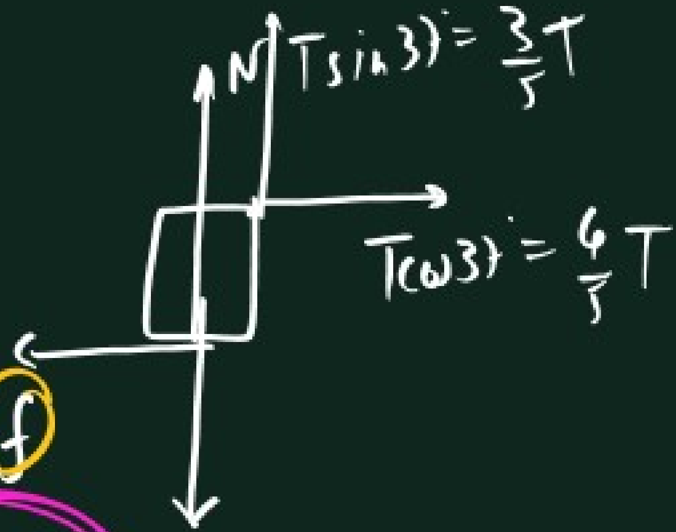
$= 36N$

5



$\mu = 0.5$

Find min. value of T to move the block.



$f_L = \mu N$ $\quad mg = 20N$

To find min. T when block just moves, $f = f_{limiting}$

$\Sigma F_y = 0$

$\Rightarrow N = 20 - \frac{3}{5}T$ — (1)

$\Sigma F_x = 0$

$\Rightarrow \frac{4}{5}T = \mu N$

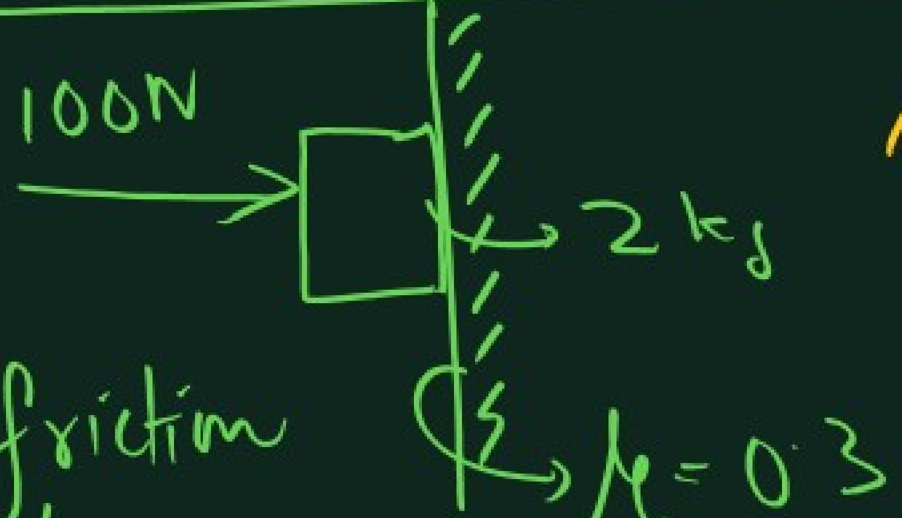
$\Rightarrow \frac{4}{5}T = \mu\left(20 - \frac{3}{5}T\right)$

$\Rightarrow \frac{4}{5}T = \frac{1}{2}\left(20 - \frac{3}{5}T\right)$

$\Rightarrow \frac{8}{5}T + \frac{3}{5}T = 20$

$\Rightarrow \boxed{T = \frac{100}{11}} \approx 9N$

6

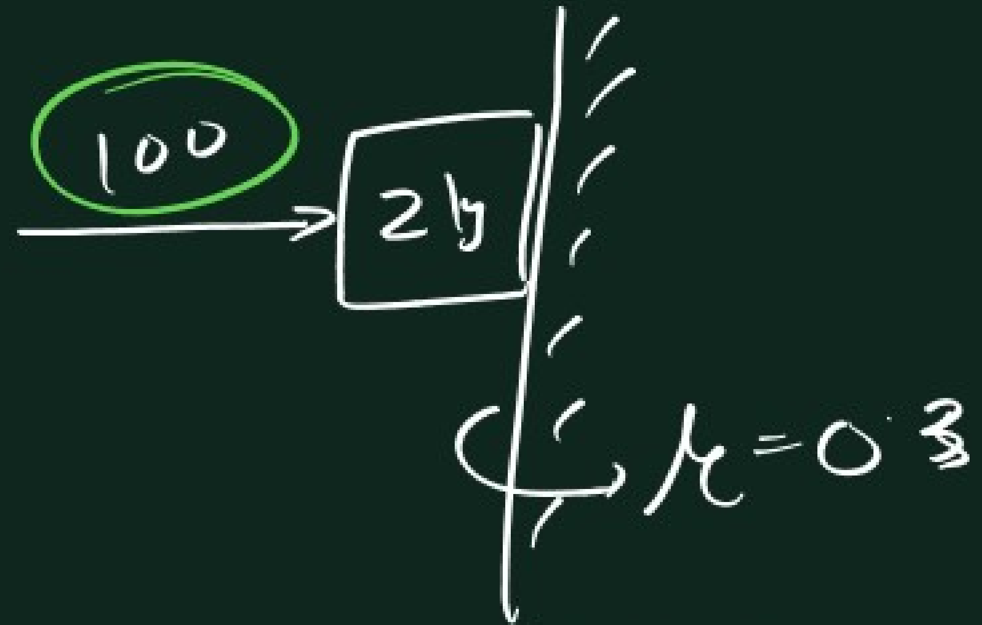


Find friction on box.

A) 30 B) 20

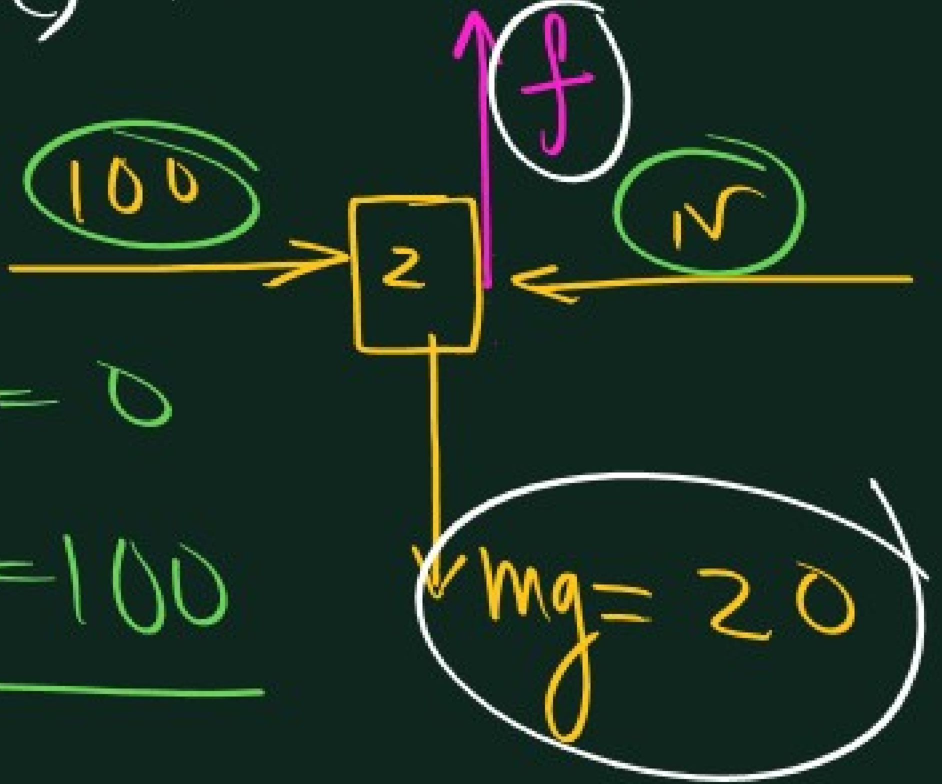
C) 10 D) None

6)



A) 30 B) 20

C) 10 D) None



$\Sigma F_x = 0$

$\Rightarrow N = 100$

$f_L = \mu N$

$= 0.3 \times 100$

$= 30N$

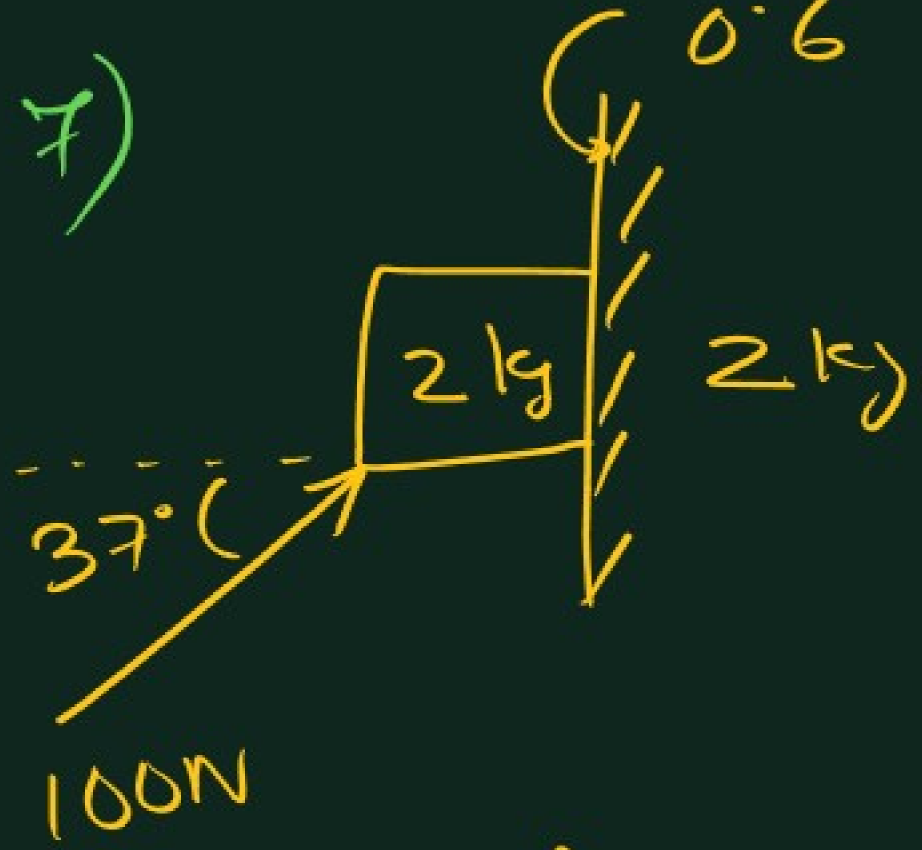
Here $\Sigma F_y = 0$

$\Rightarrow f = mg$

$\Rightarrow \boxed{f = 20N}$

$f_{acting} = 20N$

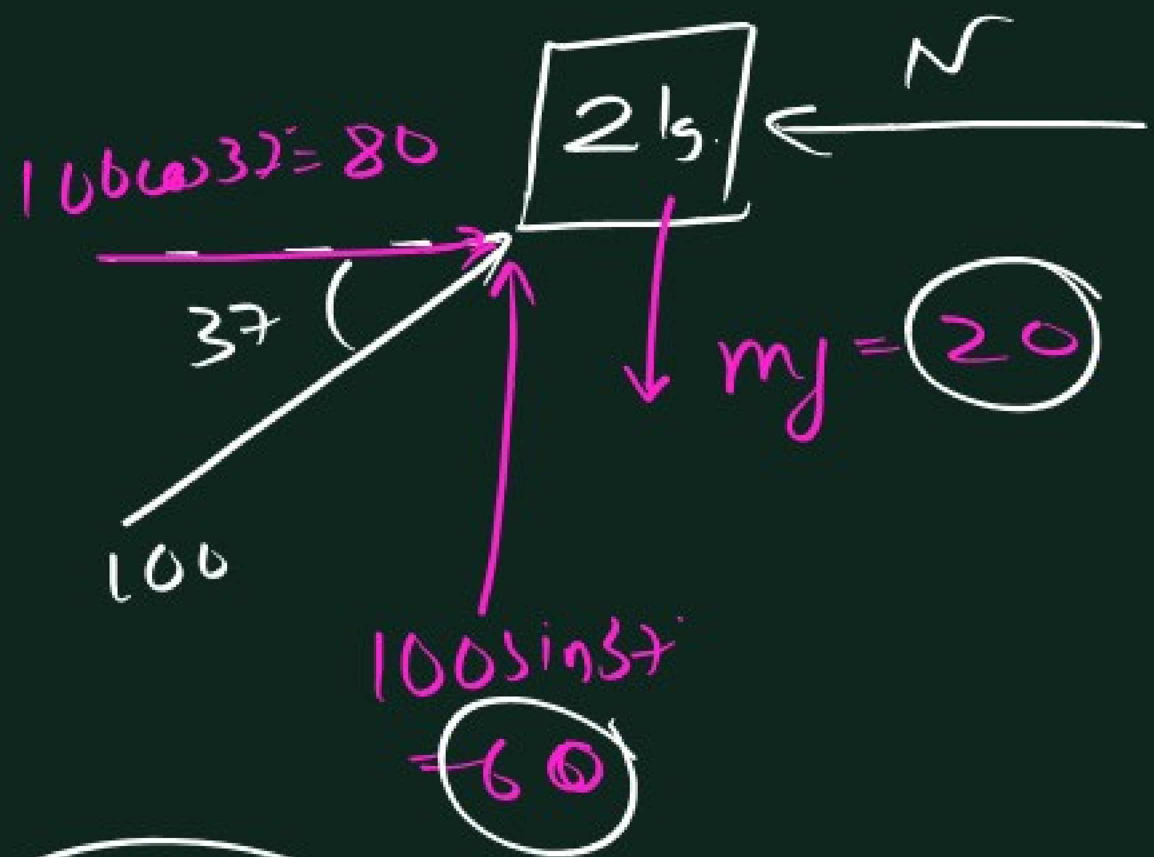
7)

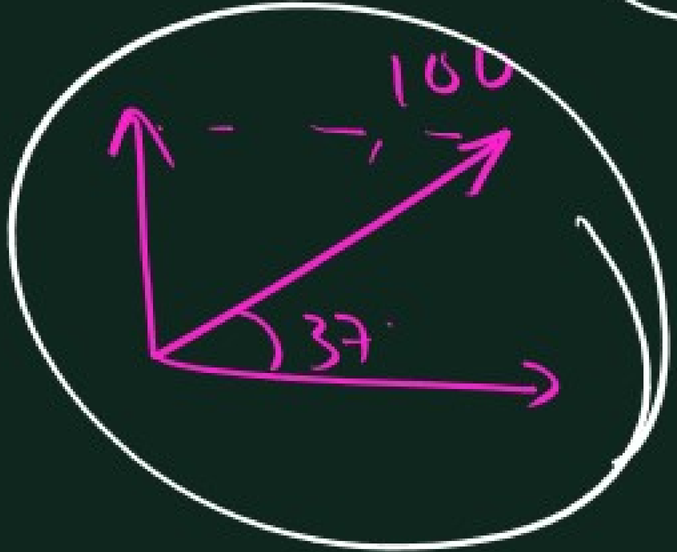


Find 1) f_L

2) f acting

7)



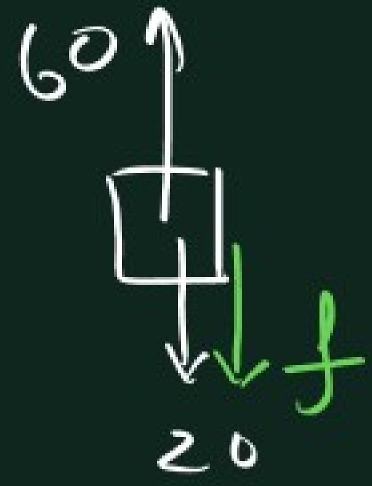


$\Sigma F_x = 0 \Rightarrow N = 80$

1) $f_k = \mu N = 0.6 \times 80$

$= 48\,N$

2)



$\Sigma F_y = 0 \Rightarrow 60 = 20 + f$

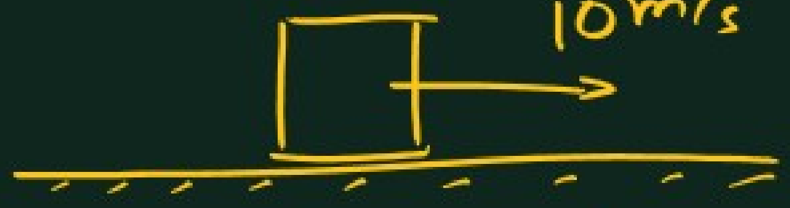
$\Rightarrow f = 40$

$\boxed{f_k = \mu_k N}$

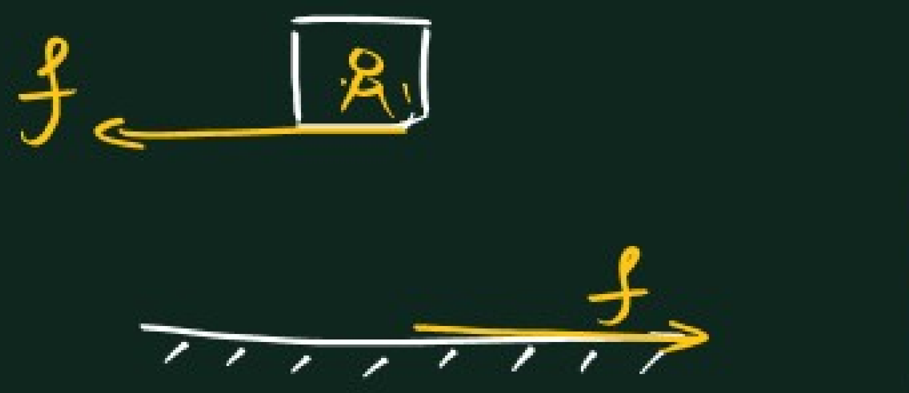
Generally

$\boxed{\mu_k < \mu_s}$

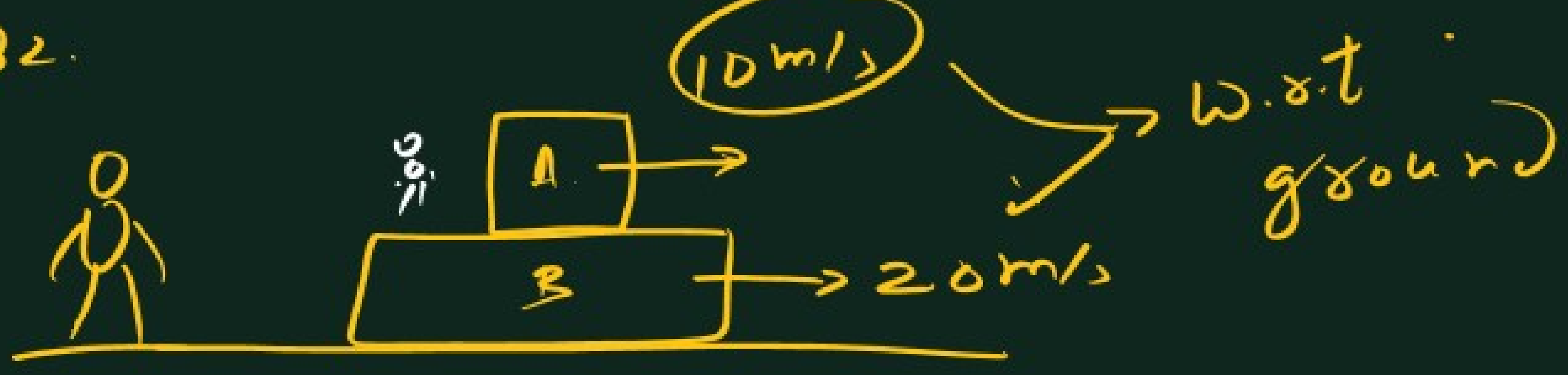
Q1.



Find direction of f on box & ground.



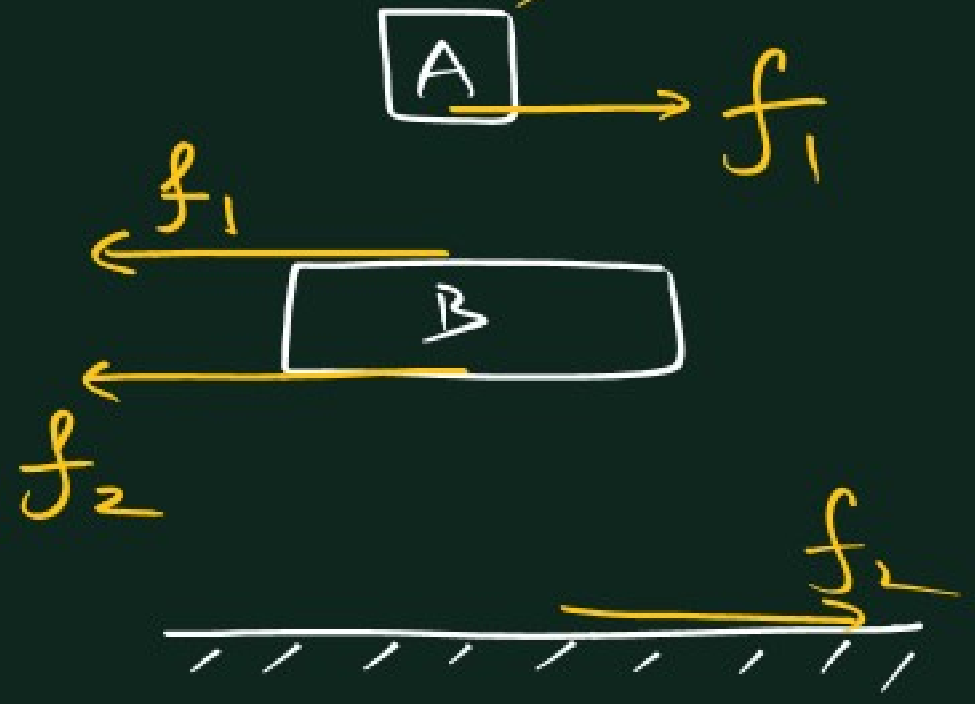
Q2.

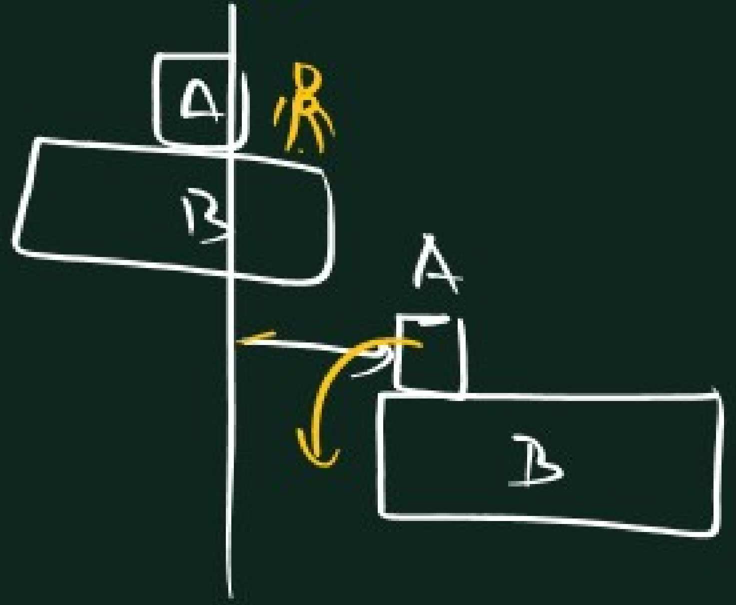


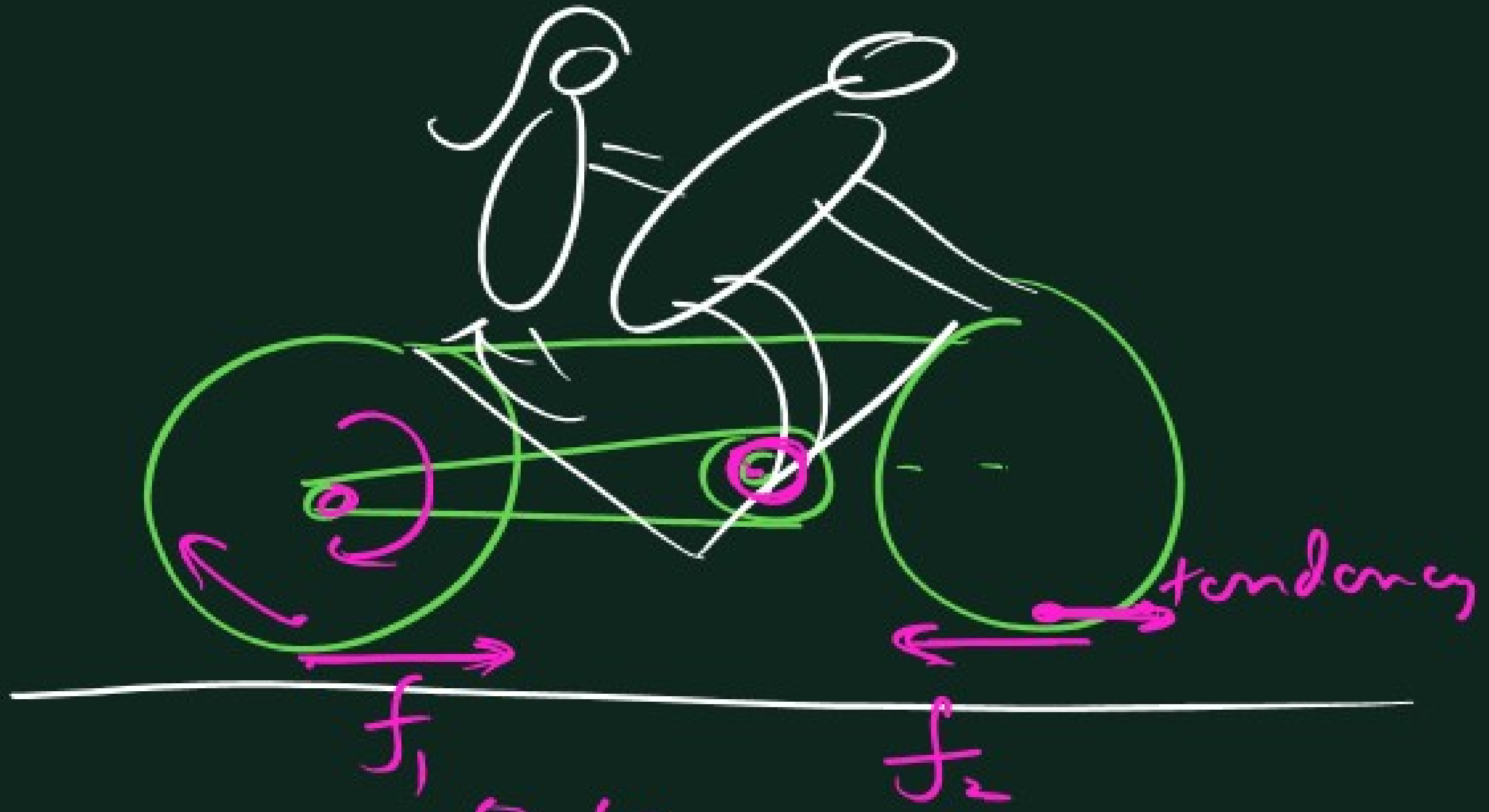
Find direction of f

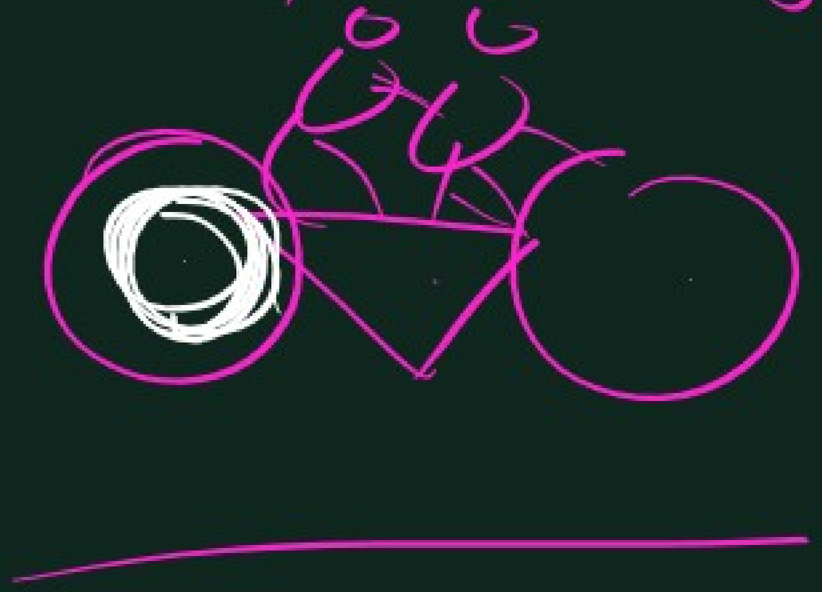
1) Between boxes

2) Between Box B & ground



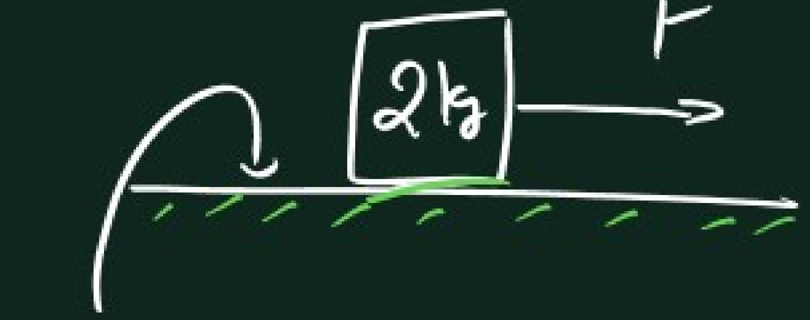






$f_1 > f_2$

Q.



$\mu_s = 0.6$

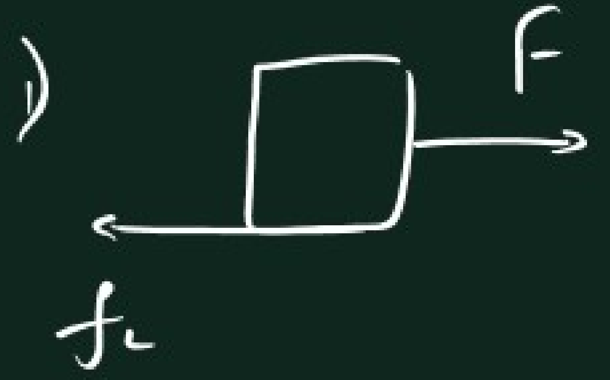
$\mu_k = 0.5$

2nd part 'a'

A) $4\ m/s^2$ B) $5\ m/s^2$

C) $6\ m/s^2$ D) $7\ m/s^2$

1)

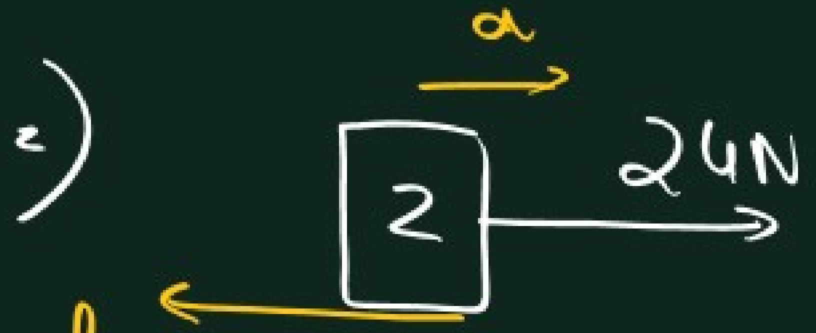


$f_L = \mu_s N = 0.6 \times 20$

$= 12N$

$\therefore F_{min} = 12N$

2)



$f_k = \mu_k N$

$f_k = \mu_k N$

$= 0.5 \times 20$

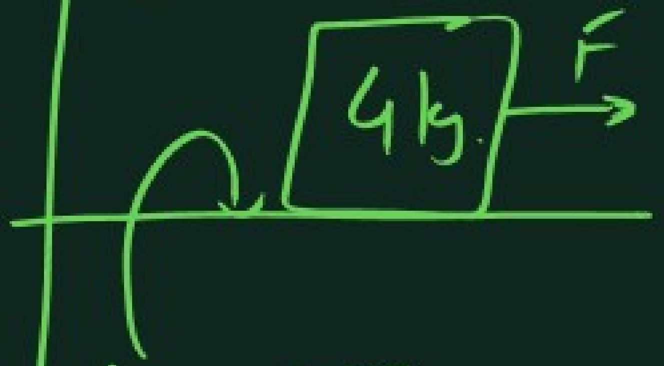
$= 10N$

$\therefore \Sigma F = ma$

$\Rightarrow 24 - 10 = 2 \times a$

$\Rightarrow \boxed{a = 7\ m/s^2}$

Q.



$\mu_s = 0.8$

$\mu_k = 0.6$

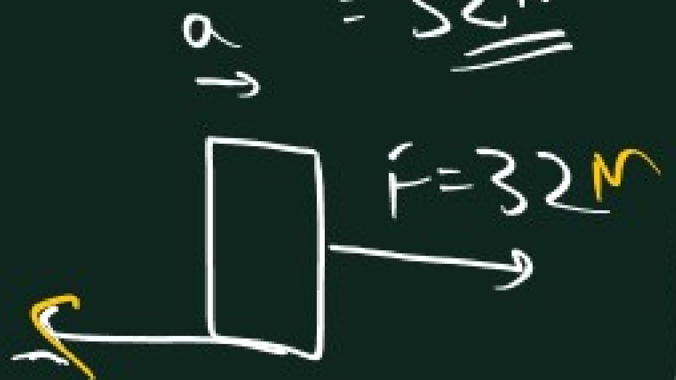
Q:

$\mu_s = 0.8,\ \mu_k = 0.6$

i) $F_{min} = f_L = \mu_s N$

$= 0.8 \times 40$

$= 32N$



$f_k = 24N$

$f_K = \mu_k N$

$= 0.6 \times 40$

$= 24N$

$\Sigma F = ma$

$\Rightarrow 32 - 24 = 4a$

$\Rightarrow 8 = 4a$

$\Rightarrow \boxed{a = 2}$

ii) $s = \cancel{ut}^{\,0} + \frac{1}{2}at^2$

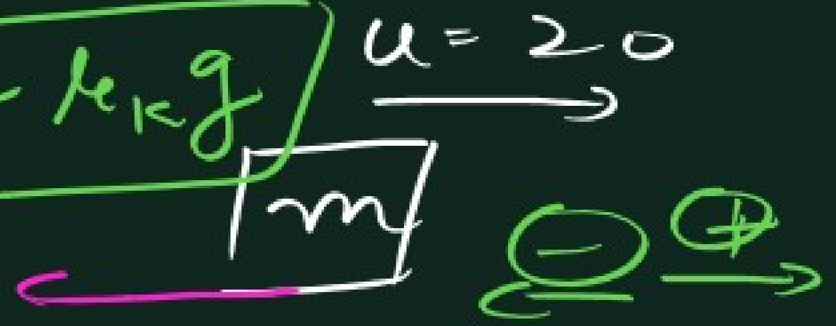
$\Rightarrow s = \frac{1}{2} \times 2 \times 10^2 = 100m$

Q.

$\mu = 0.4$

Box given a speed of 20m/s. Find distance after which it stops.

$\boxed{a = -\mu_k g}$



$f = \mu N$

$= \mu mg$

$a = \frac{f}{m} = \mu g = 4$ (0.4×10)

Sign convention

$u = 20$

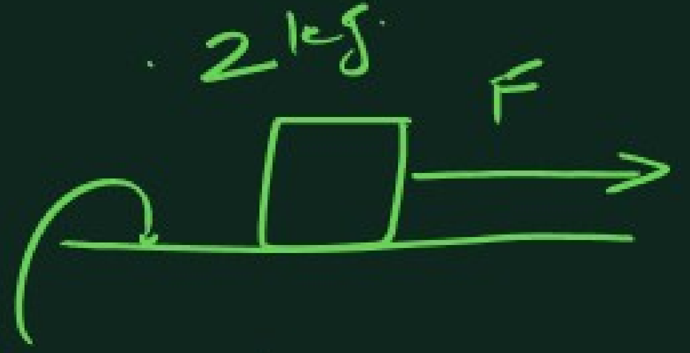
$a = -4$

$v = 0$

$s = ?!$

$v^2 = u^2 + 2as \Rightarrow \boxed{s = 50m}$

Q.



$\mu_s = 0.6$

$\mu_k = 0.4$

Find f if

i) $F = 2N, 6N, 12N, 16N, 20N$

2) Draw graph of friction(y) vs F(x)

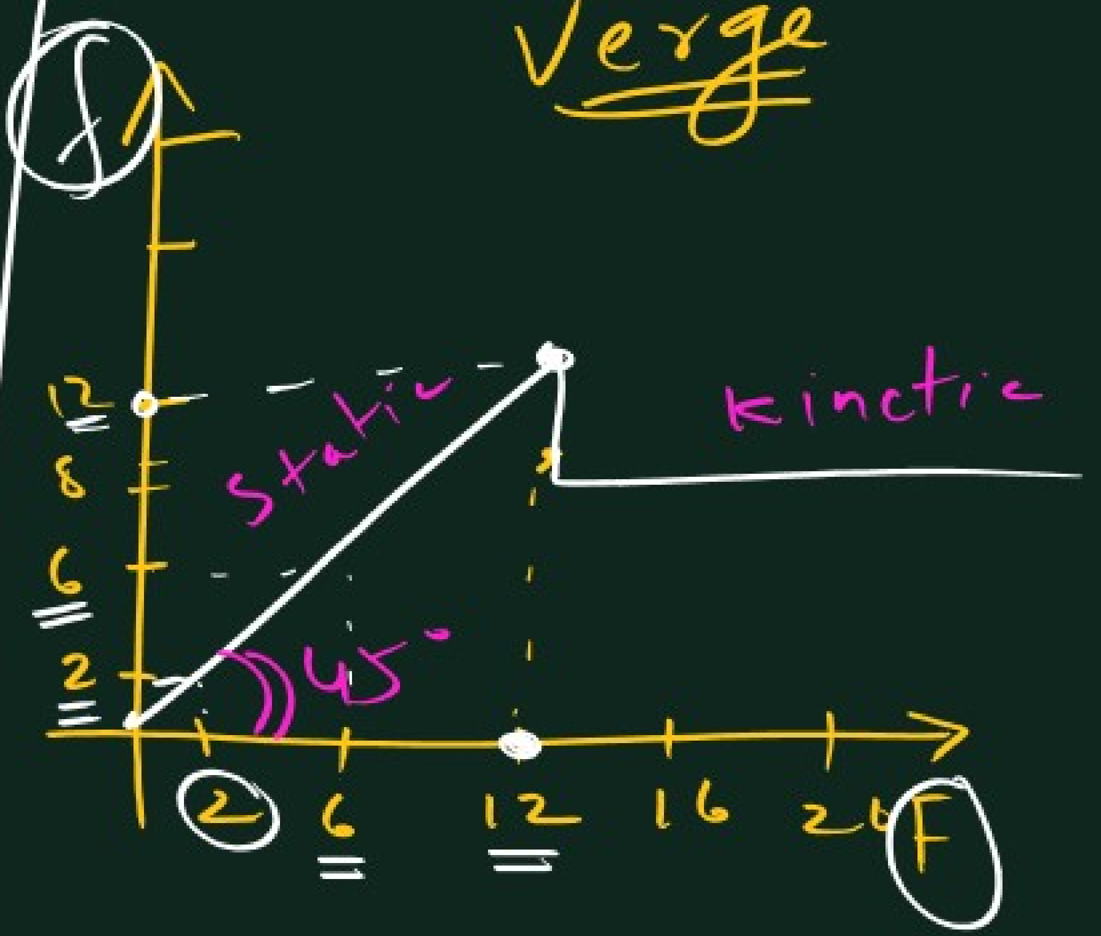
$f_L = \mu_s N = 12N$

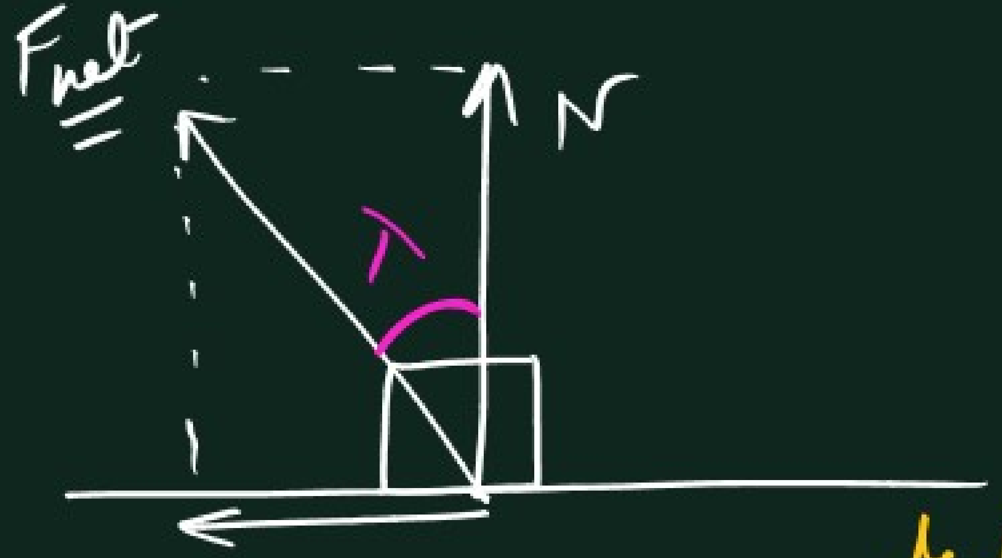
$f_k = \mu_k N = 8N$

F	f
2 →	2
6 →	6
12 →	12
16 →	8N
20 →	8N

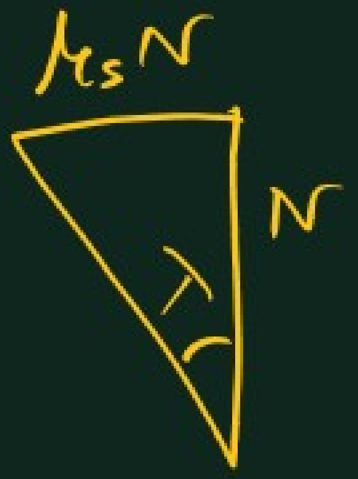
kinetic

Verge





$f_L = \mu_s N$



$\tan\lambda = \frac{\mu_s \cancel{N}}{\cancel{N}}$

$\tan\lambda = \mu_s$

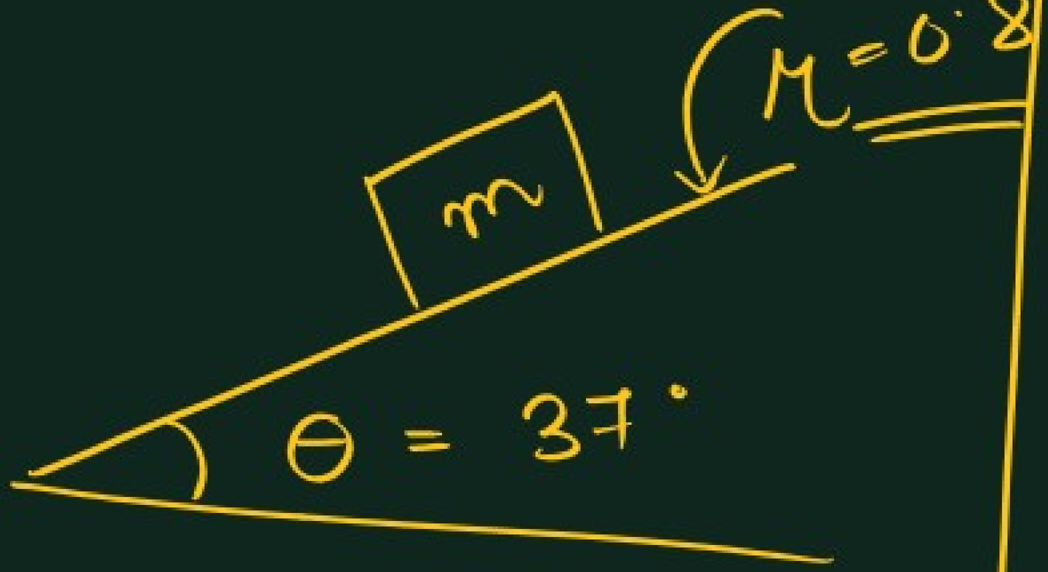
or $\boxed{\lambda = \tan^{-1}\mu_s}$

E. If angle of friction = 37°

$\Rightarrow \mu_s = \tan 37° = \frac{3}{4} = 0.75$

Friction on incline.

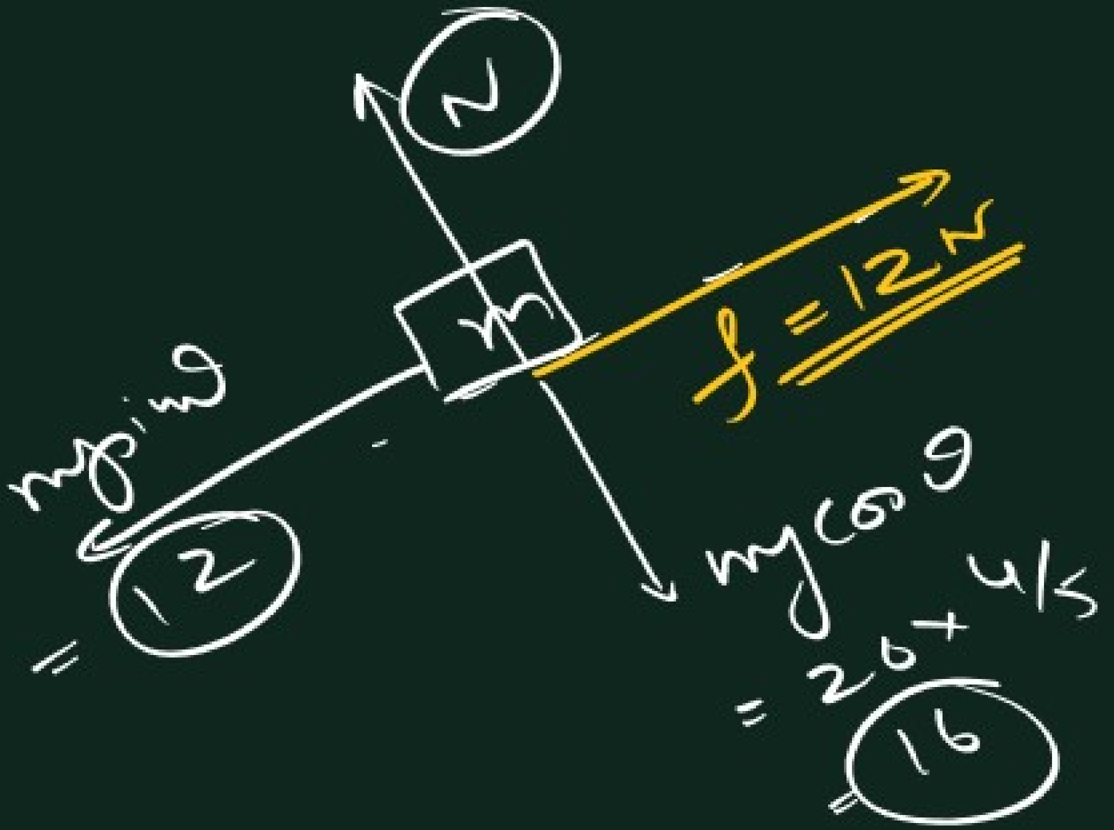
Q1.



$m = 2\ \text{kg}$.

Find f acting on block

$(g = 10\ m/s^2)$



$\sin 53 = 4/5$

$\cos 53 = 3/5$

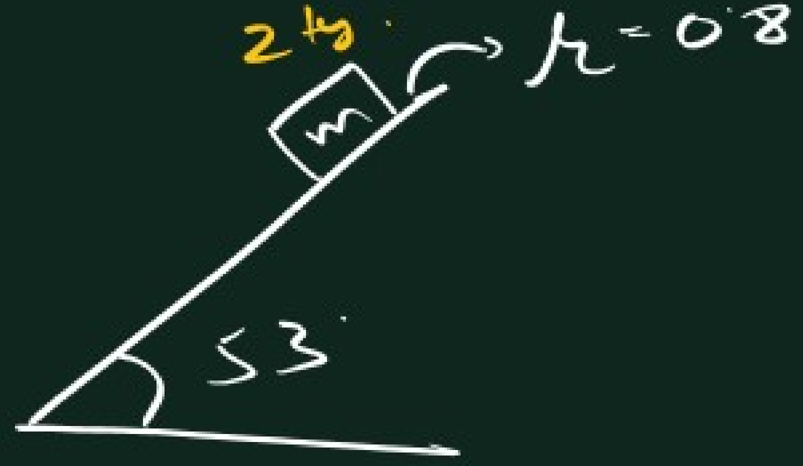
$\rightarrow \Sigma F_y = 0 \Rightarrow \underline{N = 16}$

$\rightarrow f_L = \mu_s N = 0.8 \times 16 = 12.8N$

As F_{net} parallel to surface $(mg\sin\theta) = 12N$

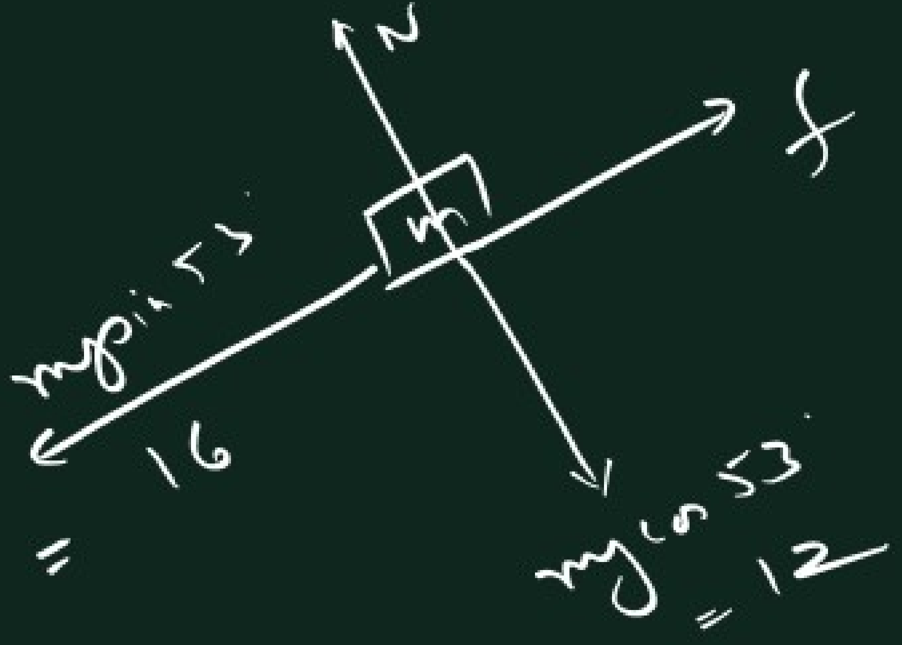
$\Rightarrow \boxed{f = 12N}$

2)



A) 12 B) 16
C) 9.6 D) 12.8

[Assume at rest
↓
Find f required
↓
Compare with $f_{Limiting}$
↓
If required $f < f_L \Rightarrow$ static f
else kinetic f]



$\Sigma F_y = 0$

$N = 12$

$\Sigma F_x = 0 \Rightarrow f = 16$

To remain at rest required $f = 16N$

$f_L = \mu_s \times N$

$= 0.8 \times 12$

$= 9.6 N$

As $f_L <$ $f_{required}$

$\Rightarrow$ slipping

$\Rightarrow$ kinetic f

$\Rightarrow f = \mu_k N$

$= 0.8 \times 12$

$= 9.6 N$

Assume object at rest

⇓

Find static f required

⇓

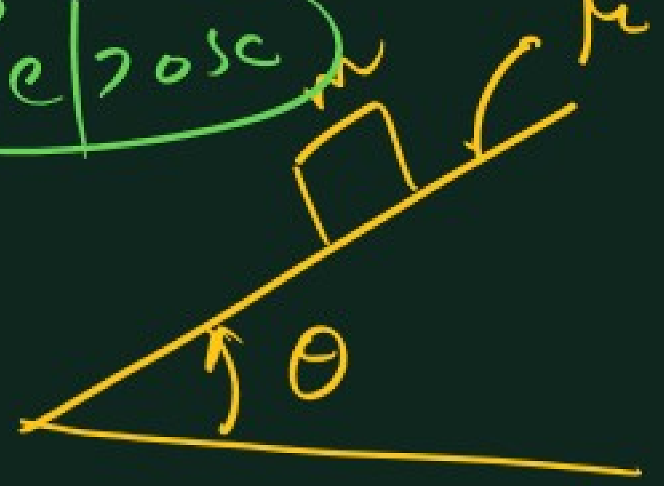
Compare with f_L (Max.)

⇓

1) { If required $f \le f_L$
object at rests, static f acts

2) { If required $f > f_L$
⇒ object slides, kinetic f acts

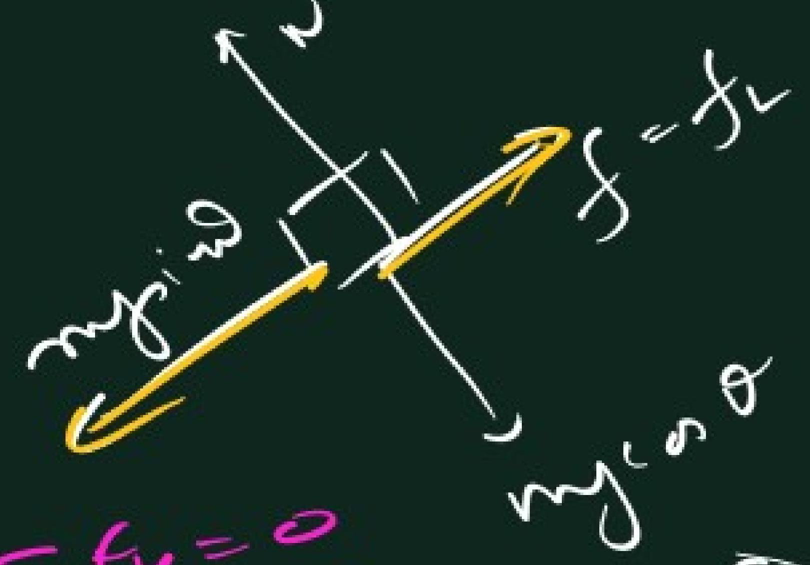
Q. Repose



If θ is gradually increased, then find the angle at which Block just slips.

A) $\tan\theta = \mu$ B) $\sin\theta = \mu$

C) $\cos\theta = \mu$ D) $\cot\theta = \mu$



$\Sigma F_y = 0$

$N = mg\cos\theta$ — (1)

$\Sigma F_x = 0$

$f_L = mg\sin\theta$

$\Rightarrow \mu N = mg\sin\theta$

$\Rightarrow \mu mg\cos\theta = mg\sin\theta$

$\Rightarrow \boxed{\tan\theta = \mu}$

$\left\{ \begin{array}{l} 3 \rightarrow \text{friction} \\ 4 \rightarrow \text{Calculus} \end{array} \right.$

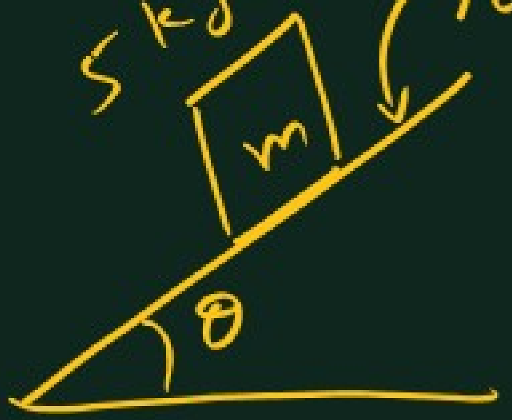
$$\boxed{\tan\theta_R = \mu_s}$$

Eg. $\mu_s = 3/4$

$\Rightarrow \tan\theta_R = 3/4$

$\Rightarrow \theta_R = 37^\circ$

Q. 5 kg. $\mu_s = 0.8,\ \mu_k = 0.7$

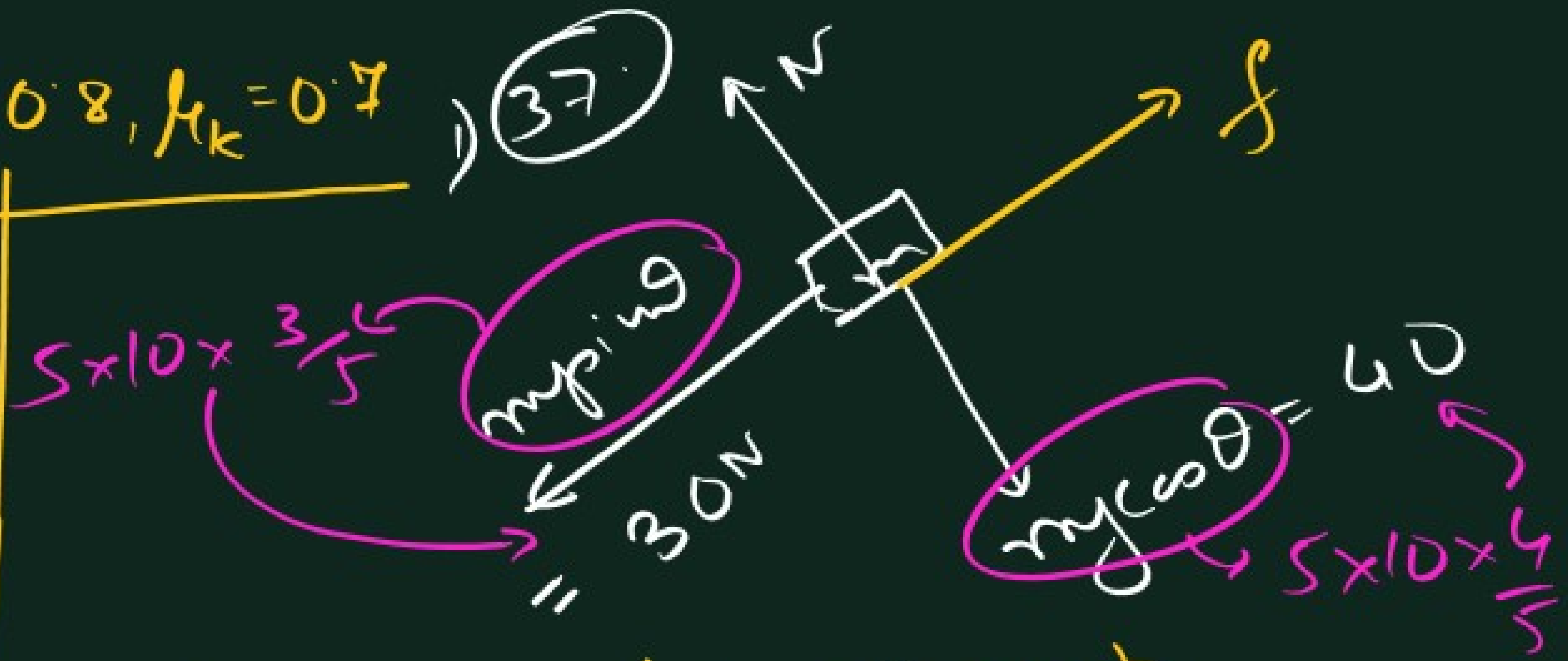


Find nature & magnitude of 'f' if

1) $\theta = 37^\circ$

2) $\theta = 53^\circ$

1) 37°



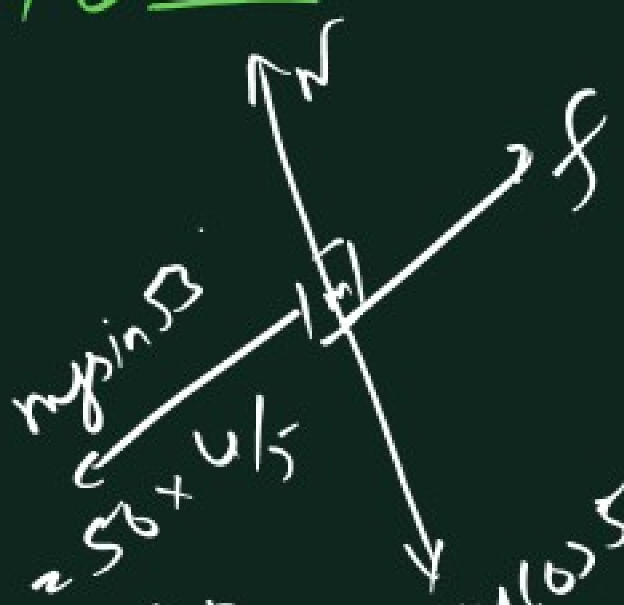
→ Assume box at rest.

$\Sigma F_x = 0 \Rightarrow f = mg\sin\theta$
$= 30N$

→ $f_L = \mu N = \mu mg\cos\theta$
$= 0.8 \times 40 = 32N$

Box at rest, static f of 30N acts.

2) $\theta = 53$



$mg\sin 53 = 50 \times 4/5 = 40$

$mg\cos 53 = 50 \times 3/5 = 30$

Assume rest

$\Downarrow$

$\Sigma F_x = 0 \Rightarrow \boxed{f = 40}$

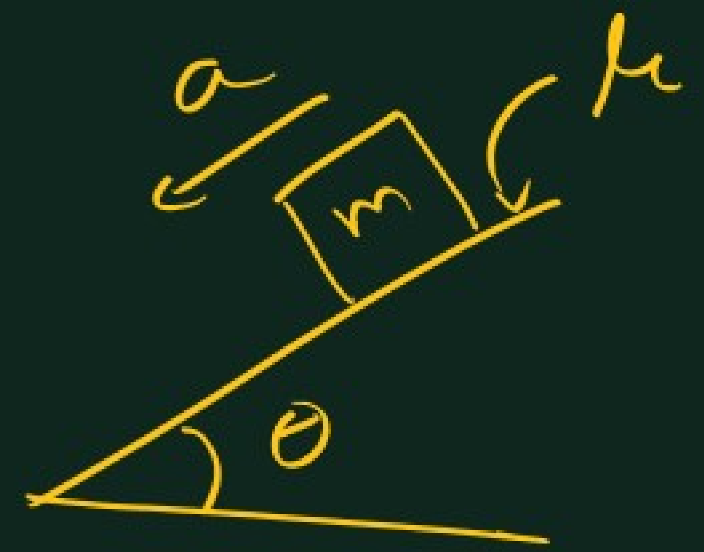
$f_L = \mu_s N = 0.8 \times 30 = 24N$

As Required $f > f_L$ (Max. possible)

$\Rightarrow$ Sliding.

$f_k = \mu_k N = 0.7 \times 30 = 21N$

Q.



If $\theta > \theta_{Repose}$

Find a of block

A) $g\sin\theta$

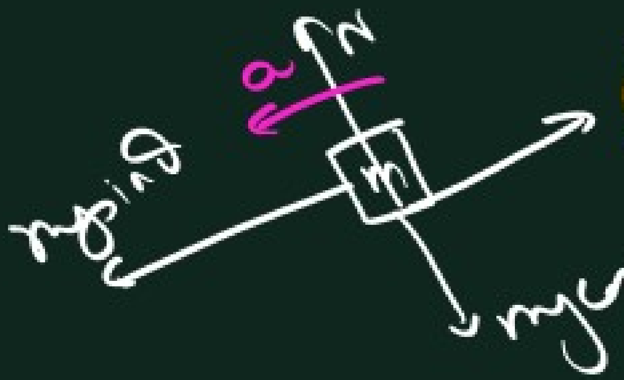
B) $\mu g\cos\theta$

C) $g\sin\theta - \mu g\cos\theta$

D) $g\sin\theta + \mu g\cos\theta$

A:- $\theta > \theta_{Repose}$

$\Rightarrow$ Sliding



$f_k \rightarrow \mu_k N$
$= \mu_k mg\cos\theta$

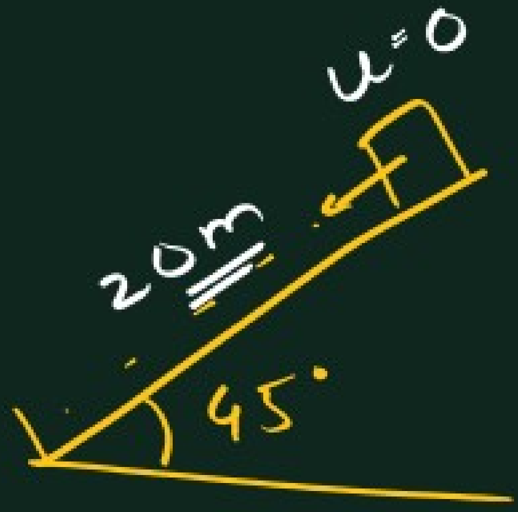
$N = mg\cos\theta$ — (1)

$\Sigma F_x = ma$

$\Rightarrow \cancel{m}g\sin\theta - \mu_k \cancel{m}g\cos\theta = \cancel{m}a$

$\Rightarrow \boxed{a = g\sin\theta - \mu_k g\cos\theta}$

Q.



Find time to slide down if $\mu_k = 3/4$.

A:- $\theta > \theta_{Repose} (37°)$

$\Rightarrow$ Slide

$a = g\sin\theta - \mu_k g\cos\theta$

$\Rightarrow a = g(\sin\theta - \mu\cos\theta)$

$= 10\left(\frac{1}{\sqrt{2}} - \frac{3}{4}\times\frac{1}{\sqrt{2}}\right)$

$= \frac{10}{\sqrt{2}} \times \frac{1}{4}$

$= \frac{5}{2\sqrt{2}}$

$\rightarrow S = \cancel{ut}^{0} + \frac{1}{2}at^2$

$\Rightarrow t = \sqrt{\frac{2s}{a}}$

$= \sqrt{\frac{2\times 20}{(5/2\sqrt{2})}}$

$= \sqrt{16\sqrt{2}}$

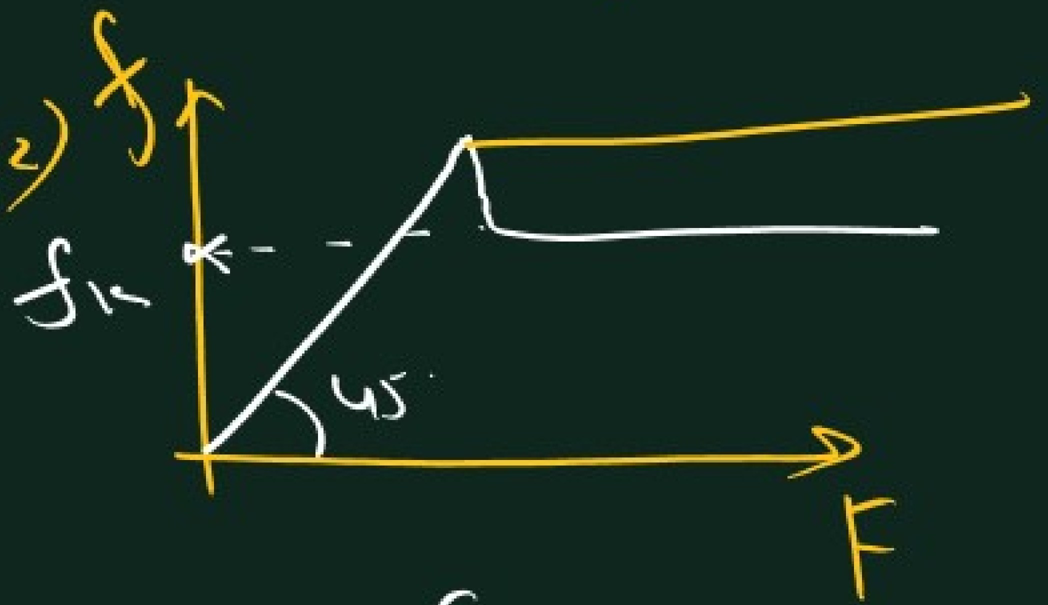
$= 4\sqrt{1.414} = 4.8\,s$

Recap

1) $0 \leq \text{Static } f \leq f_L$

$\Downarrow$

Depending on F_{ext}

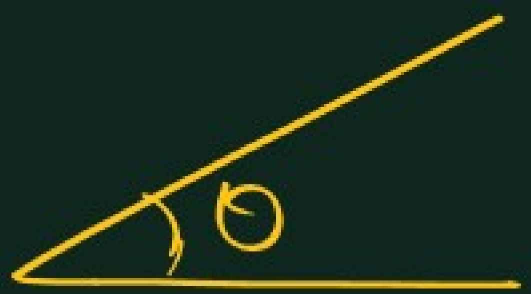
2) 

2) $f_k = \mu_k N$

$\mu_k < \mu_s$

3) 

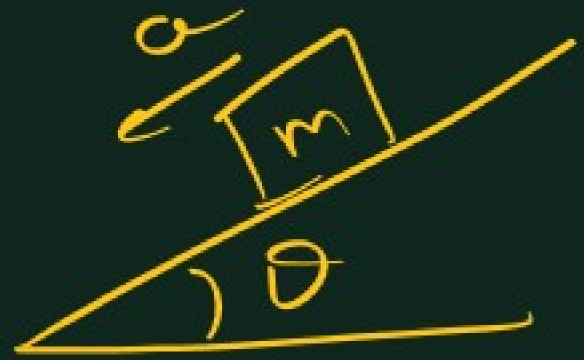
$a = -\mu g$

4) 

θ_{Repose} $\tan\theta_R = \mu$

If $\theta > \theta_R$
$\Rightarrow$ Slide

else rest.

5) 

$\theta > \theta_R$

$a = g\sin\theta - \mu g\cos\theta$

1

2. A force of 98 N is required to just start moving a body of mass 100 kg over ice. The coefficient of static friction is

(a) 0.6 (b) 0.4

(c) 0.2 (d) 0.1

2

9. The limiting friction is

(a) Always greater than the dynamic friction

(b) Always less than the dynamic friction

(c) Equal to the dynamic friction

(d) Sometimes greater and sometimes less than the dynamic friction

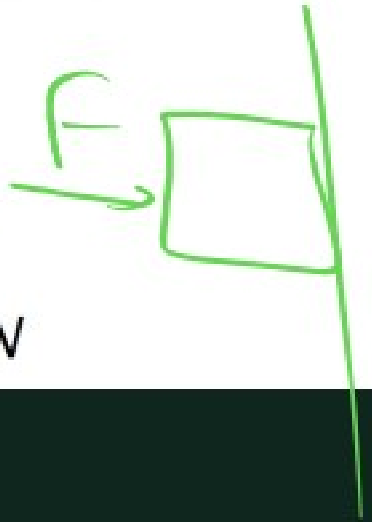
3

13. A block of 1 kg is stopped against a wall by applying a force F perpendicular to the wall. If $\mu = 0.2$ then minimum value of F will be

[MP PMT 2003]

(a) 980 N (b) 49 N

(c) 98 N (d) 490 N



4

22. A box is lying on an inclined plane what is the coefficient of static friction if the box starts sliding when an angle of inclination is 60° [KCET 2000]

(a) 1.173 (b) 1.732

(c) 2.732 (d) 1.677

$\tan\theta$

5

3. A car is moving along a straight horizontal road with a speed v_0. If the coefficient of friction between the tyres and the road is μ, the shortest distance in which the car can be stopped is [MP PET 1985; BHU 2002]

(a) $\frac{v_0^2}{2\mu g}$ (b) $\frac{v_0}{\mu g}$

(c) $\left(\frac{v_0}{\mu g}\right)^2$ (d) $\frac{v_0}{\mu}$

μmg

$\Downarrow$

μg

$$\left.\begin{aligned} u &= v_0 \\ a &= -\mu g \\ v &= 0 \end{aligned}\right\} \; v^2 = u^2 + 2as \Rightarrow \boxed{s = \frac{v_0^2}{2\mu g}}$$